HW2: Predictive Control of Dynamic System (ME 675)

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Due Date: October 5 Sunday 11:59pm

Problem 1 Minimize the following function

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 + 3|x_1 + x_2 + 2|, \quad x \in \mathcal{D} = \mathbb{R}^2$$

The minimizer can be shown to exist by the Corollary to Weierstrass theorem.

(1) Consider the set $\mathcal{D}_> = \{x : x_1 + x_2 + 2 > 0\}$. Determine whether this set is open, closed, bounded, unbounded, compact? Find local minimizer candidates if exist. Minimizer candidates must satisfy necessary conditions for optimality.

(2) Suppose now the set $\mathcal{D}_< = \{x : x_1 + x_2 + 2 < 0\}$. Determine whether this set is open, closed, bounded, unbounded, compact? Find local minimizer candidates if exist. Minimizer candidates must satisfy necessary conditions for optimality.

(3) Consider now the set $\mathcal{D}_= = \{x : x_1 + x_2 + 2 = 0\}$. For $x \in \mathcal{D}_=$, derive an expression for the one-sided (Gateaux) derivative, $d_+ f(x, u)$. Note that $f = f_1 + f_2 \Rightarrow d_+ f(x, u) = d_+ f_1(x, u) + d_+ f_2(x, u)$ if all one-sided derivatives on the right hand sides exist.

(4) Consider a point $x = [-1, -1]^T$. Is $d_+ f(x, u)$ non-negative for any direction in $\text{vcone}(\mathcal{D}, x)$?

(5) Based on your investigations above, what is the minimizer of the function?

Problem 2 Consider the following minimization problem:

$$\begin{aligned} \min_x \quad & f(x) = x_1 \\ \text{s.t.} \quad & x \in \mathcal{D} = \left\{ x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 : x_2 \le x_1^3, x_2 \ge 0, x_1 \le 1 \right\} \end{aligned}$$

(1) Sketch the set $\mathcal{D}$ on the $x_1 - x_2$ plane.

(2) Guess a minimizer, x^*. Show that your minimizer satisfies the necessary conditions $d_+ f(x^*, u) \ge 0$ for all $u \in \text{vcone}(\mathcal{D}, x^*)$. You need to determine $\text{vcone}(\mathcal{D}, x^*)$.

(3) Consider a point $x_0 = [1, 1]^T \in \mathcal{D}$. Determine $\text{vcone}(\mathcal{D}, x_0$. Are necessary conditions $d_+ f(x_0, u) \ge 0$ for all $u \in \text{vcone}(\mathcal{D}, x_0)$ satisfied?

(4) Determine the tangent cone at x_0, $\mathcal{T}(\mathcal{D}, x_0)$. Highlight the differences between $\text{vcone}(\mathcal{D}, x_0)$ and $\mathcal{T}(\mathcal{D}, x_0)$.

Problem 3 The aircraft thrust T and aircraft velocity V in steady and level flight are related by the following expression:

$$T = \frac{1}{2}\rho S V^2 C_{D0} + \frac{KW^2}{\frac{1}{2}\rho S V^2}$$

where W is the weight, ρ is the air density, S is wing surface area, and C_{D0} and K are constant parameters.

(1) Find the expression for the air-speed V of the aircraft at which the required propulsive force T is minimized. Provide also the expression for the minimum value of the propulsive thrust. You need to use the first- and second-order necessary conditions to demonstrate your result.

(2) Given that the air density experientially decreases with the increase in altitude, describe qualitatively the dependence of the optimal speed in (1) and optimal propulsive thrust on the altitude.

(3) Let $W = 73000$, $S = 950$, $C_{D0} = 0.015$, $K = 0.05$, $\rho = 1.7553 \times 10^{-3}$. Substitute these numerical values into the expression in (1) and give the numerical values for the optimal air-speed and minimum value of the propulsive thrust.

(4) Generate a Matlab plot of the propulsive thrust vs. air-speed and confirm your result in (4). Consider $V \in [100, 900]$ in generating your plot.

Problem 4 Explain why the following functions are convex

(1) $f = x$, $x \in \mathcal{R}$.

(2) $f = -x$, $x \in \mathcal{R}$.

(3) $f = |x|$, $x \in \mathcal{R}$.

(4) $f = x^2$, $x \in \mathcal{R}$.

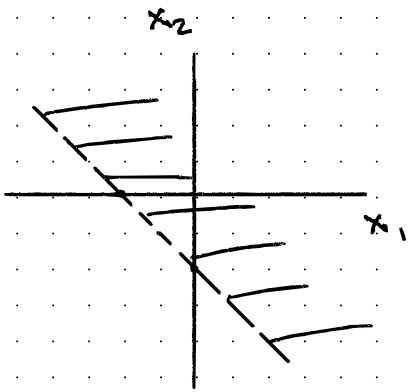
(5) $f = x^2 + |x|$, $x \in \mathcal{R}$.

(6) $f = x_1^2 + x_2^2 + 2\max\{x_1, x_2\}$, $x \in \mathcal{R}^2$.

Problem 1

$$f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 + 3|x_1 + x_2 + 2|, \quad x \in D = \mathbb{R}^2$$

1) $D_> = \{x : x_1 + x_2 + 2 > 0\}$



Set $D_>$ is open & unbounded, <u>not</u> compact.

$$\nabla f = \left(\frac{\partial}{\partial x_1}\hat{i} + \frac{\partial}{\partial x_2}\hat{k}\right) f = 0$$

$$= \begin{bmatrix} 2x_1 + x_2 + 3 \\ x_1 + 2x_2 + 3 \end{bmatrix} = 0$$

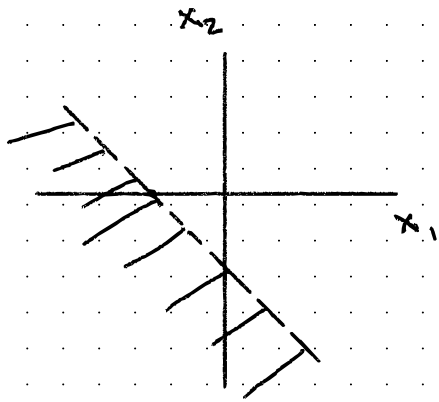
$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \notin D_>$$

No local minimizer candidate

2) $D_< = \{x : x_1 + x_2 + 2 > 0\}$

x_2

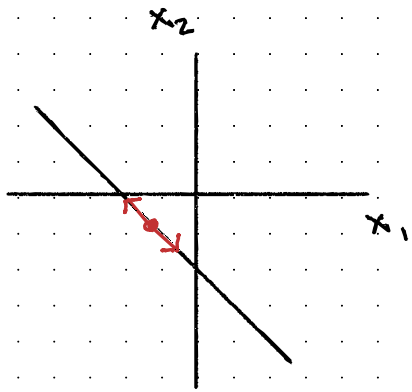
Set $D_<$ is open & unbounded, <u>not</u> compact.



$x = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \notin D_<$

No local minimizer candidate

3) $D_= = \{x : x_1 + x_2 + 2 = 0\}$



set $D_=$ is closed but unbounded, not compact

$x = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \in D$

$-1 + -1 + 2 = 0$

Candidate for (local) minimizer

$d_+ f(x; u) = \nabla f^T u \qquad \nabla f = \begin{bmatrix} 2x_1 + x_2 + 3 \\ x_1 + 2x_2 + 3 \end{bmatrix}$

$d_+ f(x; u) = \begin{bmatrix} 2x_1 + x_2 + 3 & x_1 + 2x_2 + 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$$\boxed{d_+ f(x; u) = u_1(2x_1 + x_2 + 3) + u_2(x_1 + 2x_2 + 3)}$$

4)

$\text{vcone}(D, x) = \left\{ u \in \mathbb{R}^2 \,\middle|\, \frac{u_2}{u_1} = -0.5 \right\}$

$d_+ f\left(x_0 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}; u\right) = u_1(-2 - 1 + 3) + u_2(-1 - 2 - 3)$

$= 0 \quad \forall u$

Non-negative for any vector $u \in \text{vcone}(D, x_0)$

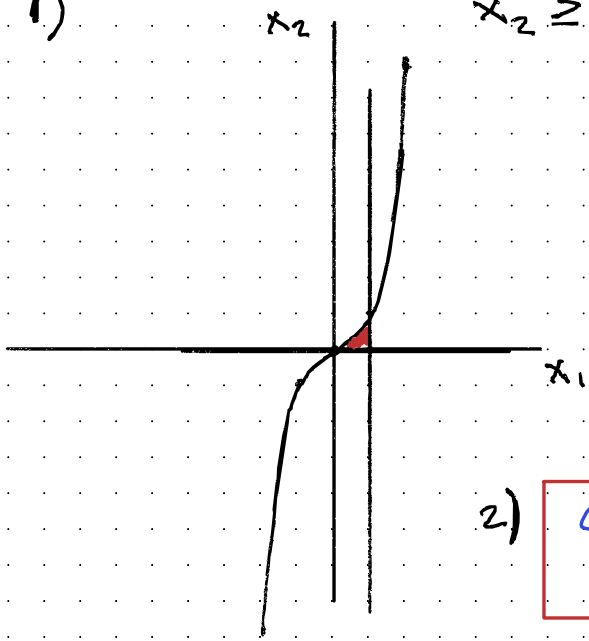
5) $\boxed{x = [1 \ 1]^T \text{ is minimizer}}$

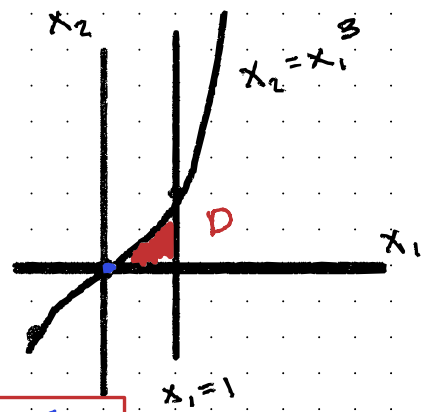
Problem 2

$$\min_x f(x) = x_1$$

$$\text{s.t.} \quad x \in D = \left\{ x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 : x_2 \le x_1^3, \; x_2 \ge 0, \; x_1 \le 1 \right\}$$

1)



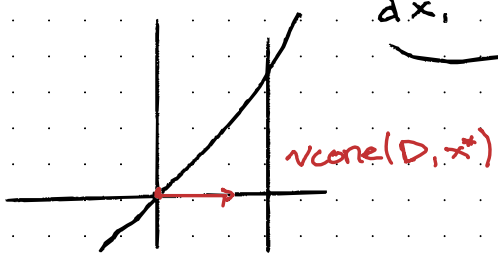


2) $\boxed{\text{guess, } x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}}$

$$d_+ f(x^*; u) = \Delta f(x^*)^T u \qquad u \in \text{vcone}(D, x^*)$$

$$\text{vcone}(D, x^*) = \{ u \in \mathbb{R} \mid u_1 = [0, \infty), u_2 = 0 \}$$

$$x_2 = x_1^3, \quad \underbrace{\frac{d x_2}{d x_1} = 3x_1^2 \Big|_{x_1 = 0} = 0}_{u_2 = 0}$$



$$\nabla f = \left\langle \frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2} \right\rangle f = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$d_+ f(x^*, u) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = u_1 + 0$$

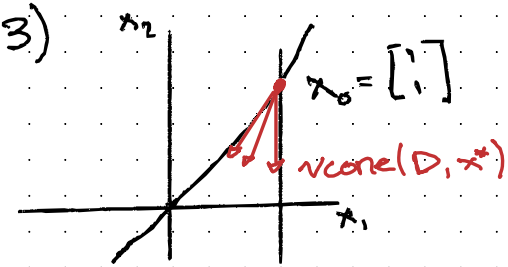
always non-negative

$[0, \infty)$

$$\text{vcone}(D, x^*) = \{ u \in \mathbb{R} \mid u_1 = [0, \infty), u_2 = 0 \}$$

$$\boxed{\text{Necessary condition } d_+ f(x^*, u) \geq 0 \text{ is satisfied}}$$

3)



$$\left.\frac{d x_2}{d x_1}\right|_{x = x_0} = \left. 3 x_1^2 \right|_{x = x_0} = 3$$

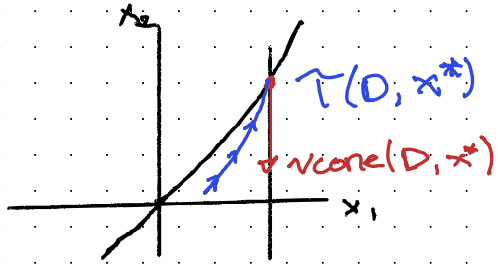
$$\text{vcone}(D, x_0) = \left\{ u \in \mathbb{R}^2 \mid u_2 = (-\infty, 0], \frac{u_2}{u_1} \leq 3 \right\}$$

$$d_+ f(x_0; u) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \underbrace{-3 + 0 < 0}$$

example u

$$\boxed{\text{Necessary cond. for first order FAILS}}$$

4)



$$\left.\frac{dx_2}{dx_1}\right|_{\begin{bmatrix}1\\1\end{bmatrix}} = 3$$

$$T(D, x_0) = \{u \in \mathbb{R}^2 \mid u_1 \leq 0,\ u_2 \leq 3u_1\}$$

$$vcone(D, x_0) = \{u \in \mathbb{R}^2 \mid u_2 = (-\infty, 0],\ \frac{u_2}{u_1} \leq 3\}$$

$$\boxed{T(D, x_0) = vcone(D, x^*)}$$

→ In this case, tangent cone & variational cone describe the same vector space.

115

176

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Problem 3

1) Expression for air speed V where thrust T is minimized

$$T = \frac{1}{2}\rho S V^2 C_{p_0} + \frac{2KW^2}{\rho S V^2}$$

$$\frac{dT}{dV} = (\rho S C_{p_0}) V - 2\left(\frac{2KW^2}{\rho S}\right) V^{-3}$$

$$\frac{dT}{dV} = (\rho S C_{p_0}) V - \left(\frac{4KW^2}{\rho S}\right) V^{-3} = 0$$

$$\boxed{V^* = \left(\frac{4KW^2}{\rho^2 S^2 C}\right)^{1/4}}$$ ← Minimizer candidate

$$\frac{d^2T}{dV^2} = \rho S C_{p_0} + \frac{12KW^2}{S V^4 \rho}$$

@ V^* ↘

$$\left.\frac{d^2T}{dV^2}\right|_{V=V^*} = 4 C S_{p_0} \rho > 0$$

≥ 0 $(C, S_{p_0}, \rho \in \mathbb{R}^+)$

↑ > 0, so V^* is a minimizer

Lets find T^* for the minimizer V^*

$$T^* = T\Big|_{V=V^*}$$

$$\boxed{T^* = \frac{2KW^2}{S\rho\sqrt{\frac{KW^2}{CS^2\rho^2}}}}$$

2) Optimal Air Speed

$$\boxed{V^* = \left(\frac{4KW^2}{\rho^2 S^2 C}\right)^{1/4}}$$

- $V^* \propto \frac{1}{\rho^{1/2}}$
- ↑ Altitude, ↓ ρ, ↑ V^*
- Optimal air speed <u>inc.</u> w/ altitude

Optimal Thrust

$$\boxed{T^* = \frac{2KW^2}{S\rho\sqrt{\frac{KW^2}{CS^2\rho^2}}}}$$

- $T^* \propto \frac{1}{\rho\sqrt{\rho^{1/2}}} = \frac{1}{\rho^{5/4}}$
- ↑ Altitude, ↓ ρ, ↑ T^*
- Optimal Air Speed T^* increases w/ Altitude

3) $T^* = 3998.4$

$V^* = 399.8$

4)